Problemes 1. MATRIUS

1.1

Dadas las siguientes matrices:

$$A = \begin{pmatrix} 3 & 0 \\ -1 & 5 \end{pmatrix}; \qquad B = \begin{pmatrix} 4 & -2 & 1 \\ 0 & 2 & 3 \end{pmatrix}; \qquad C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix};$$

$$D = \begin{pmatrix} 0 & -3 \\ -2 & 1 \end{pmatrix}; \qquad E = (4, 2); \qquad F = \begin{pmatrix} -1 \\ 2 \end{pmatrix};$$

realizar, si es posible, las siguientes operaciones:

1. $3D - 2A$.
2. $B - C^t$.
3. $D + BC$.
4. $B^t B$.
5. EAF.
6. $B^t C^t - (CB)^t$.

1.2

Si

$$A = \begin{pmatrix} 1 & 4 \\ -3 & 1 \end{pmatrix},$$

comprobar que $A^2 - 2A + 13I_2 = 0$ y utilizar esta expresión para deducir el valor de A^{-1}.

1.3

Si A es idempotente, probar que $B = I - A$ también lo es.

1.4

Probar que si A y B son matrices ortogonales cuadradas (es decir, que $A^{-1} = A^t$ y $B^{-1} = B^t$) y el producto AB existe, entonces AB es también ortogonal.

1.5 Mostrar con un ejemplo que $A\,B = C\,B$ no implica que $A = C$. ¿En qué caso sí se verifica necesariamente?

1.6 Sea A una matriz $n \times 1$ (vector columna) tal que $A^t A = 1$. Construimos las matrices $B = I_n - A\,A^t$ y $C = I_n - 2\,A\,A^t$. Probar que B es idempotente y que $C^2 = I_n$.

1.7 Probar que si $A \in \mathbb{K}^{m\times n}$ y $B \in \mathbb{K}^{n\times m}$, entonces $\operatorname{tr}(A\,B) = \operatorname{tr}(B\,A)$.

1.8 Suppose the first two columns, $\mathbf{b}_1$ and $\mathbf{b}_2$, of B are equal. What can you say about the columns of AB (if AB is defined)? Why?

1.9 Suppose the third column of B is the sum of the first two columns. What can you say about the third column of AB? Why?

1.10 Suppose the second column of B is all zeros. What can you say about the second column of AB?

1.11 Suppose the last column of AB is entirely zero but B itself has no column of zeros. What can you say about the columns of A?

1.12 Show that if the columns of B are linearly dependent, then so are the columns of AB.

1.13 If (a, b) is a multiple of (c, d) with $abcd \neq 0$, *show that* (a, c) *is a multiple of* (b, d). This is surprisingly important; two columns are falling on one line. You could use numbers first to see how a, b, c, d are related. The question will lead to:

If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has dependent rows, then it also has dependent columns.

1.14 *True or false*

(a) If the 5 by 2 matrices A and B have independent columns, so does $A + B$.

(b) If the m by n matrix A has independent columns, then $m \geq n$.

(c) A random 3 by 3 matrix almost surely has independent columns.

1.15 If A and B have rank 1, what are the possible ranks of $A + B$? Give an example of each possibility.

1.16 Demuestre, mediante inducción matemática, que

$$A^n = \begin{bmatrix} \cos n\theta & -\operatorname{sen} n\theta \\ \operatorname{sen} n\theta & \cos n\theta \end{bmatrix} \text{ para } n \geq 1$$

1.17 For the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, find A^2, A^3, A^4. Find a general formula for A^n for any positive integer n.

1.18 For the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, find A^2, A^3, A^4. Find a general formula for A^n for any positive integer n.

1.19 **(a)** Proporcione un ejemplo para demostrar que si A y B son matrices simétricas de $n \times n$, entonces AB no necesita ser simétrica.

(b) Demuestre que, si A y B son matrices simétricas de $n \times n$, entonces AB es simétrica si y sólo si $AB = BA$.

1.20 Si A y B son matrices de $n \times n$ antisimétricas, ¿en qué condiciones AB es antisimétrica?